

The Effect of Financial Literacy and Fintech Knowledge on Fintech Services Usage

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Abstract—This research aims to observe the effect of financial literacy on fintech service usage with the support of fintech knowledge. Data collecting through a questionnaire with 210 students from BINUS Business School, Bina Nusantara University. The questionnaire was made using a Likert scale. The collected data is then processed using the structural equation modelling method. Indicators measure the financial literacy variable: understanding interest rates, understanding inflation, financial skills, financial attitude, management of finances, and risk and return in investing. The fintech knowledge variable is measured by indicators, namely knowledge about fintech practices, knowledge about fintech terms, knowledge from college, familiarity in fintech, and awareness of fintech. The fintech services usage variable is measured by indicators of access features, service reliability, user's needs, user friendly application, trust, and intention. The study results show that financial literacy has significant and positive influence on fintech knowledge. Financial literacy has significant and positive influence on fintech services usage with the support of fintech knowledge.

Keywords—financial literacy, fintech knowledge, fintech services usage, financial technology, structural equation modelling

I. INTRODUCTION

The term fintech has appeared in various business journals to introduce financial services that are cheaper, faster, and human-centred. The visionary statement from 1994 made by Bill Gates stated about necessary of banking but not necessary for banks, where this statement has become a prediction that is getting stronger with the development of fintech companies around the world [1].

A new era of fintech is about to begin; This phenomenon covers various sectors such as finance, technology and innovation. The word fintech arises from combining two terms, namely finance and technology. Fintech describes the relationship of multiple technologies with financial assets. Fintech represents innovative technology-based methods of performing activities directly or indirectly connected to financial services [2].

Fintech can describe any innovation associated to the businesses technique through develop the process, distribution, and using financial services [3]. Currently, fintech is quickly becoming a global phenomenon. The rapid development of fintech is supported by increasingly affordable mobile phone and computer prices and technology advances in internet services that can reach more and more users at a low cost, making people who use fintech services increase.

The challenge for fintech companies in developing countries is individuals who have a limited understanding of basic financial concepts or individuals who are new to informal types of loans. In essence, fintech products and services must provide suitable accessibility and ease of use. Eventually, for successful fintech in typically developing countries, fintech service providers is necessity to familiarize the company with unique and diverse customers. The understanding of problematic when facing potential customers and have local knowledge about their level of financial literacy is important factor to the efficacious design of fintech services and implementation in developing countries [4].

As one of the developing countries, fintech in Indonesia has become a rapidly growing industry in the field of financial services, the services provided by fintech companies represent companies that provide services traditionally such as financial institutions and banks. Government of Indonesia in 2017 published the E-Commerce Road Map. The regulation on E-Commerce Road Map is issued considering that the economy electronic-based has good potential for Indonesia economics. Economy electronic-based can be the backbones and support national economy. To improve the utilization of potential economy electronic-based, the government needed to encourage the acceleration and development of national trading systems, start-ups, business advance, and improvement of logistics by creating an integrated Road Map of E-Commerce (Website Ministry of Communication and Information Technology, www.kominfo.go.id).

The development of the fintech industry through the E-Commerce Road Map needs to be supported by improving financial literacy in the community and financial literacy education in students. Financial literacy in students is essential to support their future success. Currently, knowledge about the financial sector is low; this is in line with access to financial institutions that are still inferior compared to other countries. This financial access has an important role in improving the community's standard of living or welfare.

The government is aggressively making various efforts to raise the level of public understanding of the financial sector and access to the financial sector. Based on the surveys carried out in 2019 by the Financial Services Authority regarding Financial Literacy and Inclusion, financial literacy and financial inclusion levels reached 38.03% and 76.19%, respectively. The target set by the government for financial literacy level is 35%, while the financial inclusion level is 75%. These achievements must improve to support the

National Economic Recovery Program and achieve the financial inclusion target of 90% by the end of 2024 (Website of the Indonesian Financial Services Authority, www.ojk.go.id).

Research on Knowledge of Fintech in Pakistan highlights that most students believe that the academic syllabus they obtain does not adequately address the content regarding fintech. Students also think of the need to understand and gain the proper knowledge about fintech. The majority of students show consensus to take courses on fintech, which shows that students consider it important to know fintech, especially in the current era of digitalization [5]. Knowledge of fintech characteristics would assist customers in making decisions concerning the appropriate service to use and know that fintech offers simple and easier procedures for credit application [6].

Program of financial education should have concentration to increase financial knowledge objective amid young people. Various teaching approaches enhance students' subjective knowledge, and improved education of financial literacy will ensure students skill development related financial behaviors. Instructors need to have attention about the student knowledge limitations and include more practice oriented educational programs to inspire students to involve in confident financial behavior through educational process [7].

The rapidly growing fintech industry and high student interest in using Fintech services have encouraged researchers to research financial literacy that can affect fintech knowledge in students and its impact on fintech usage services. This research is expected to contribute to developing studies related to fintech usage services among students.

Based on the explanations that have been given before, researchers conducted a study of financial literacy phenomena that affect fintech knowledge and its impact on fintech services usage among students. This research contributes to encouraging the use of fintech applications among students with the support of knowledge about fintech and finance.

II. LITERATURE REVIEW

The development of Fintech is currently very rapid, supported by computer technology, the internet and mobile phone that is increasingly advanced. Fintech is one of the most promising industries determined by a wave of innovative start-ups with new business models, new products and services, and new revenue models changing for better financial services worldwide [8]. Definition Fintech [9] innovation which is in financial services by means of new technology tools, for instance mobile technology, cloud computing, and big data. Fintech is divided into eight industrial sectors [10]: money transfers and payments; fintech lending; digital banking; capital markets innovations; digital wealth managers; InsureTech (insurance); equity crowdfunding; PropTech (property).

Financial literacy definition [11] is combination of skill, awareness, knowledge, behavior, and attitude required to make wide-ranging financial decisions and accomplish individual economic welfare [11]. According to the Indonesian Financial Services Authority [12], long-term goals from financial literacy for all groups of society, that is

increasing someone literacy who was formerly not literate or less literate to well literate and increasing the amount of users of financial products and financial services. Financial literacy delivers excellent benefits for industrial sectors of financial services; the developed the community's financial literacy level, the more society will use financial products and financial services.

Financial literacy includes knowledge, skills, behaviour and attitudes. The importance of applying financial literacy in real-life financial decision-making impacts economic well-being [13]. Identification of other factors that play a crucial role and add to the importance of approaches in financial literacy, namely financial attitudes and financial skills, are significant determinants and predictors of financial literacy levels [14]. The financial literacy measurement involves three parts: financial knowledge, financial behavior, and attitudes towards financial planning.

A set of perceptions of financial literacy focuses on personal finance or money management, debt, savings, investments, insurance and pensions [16]. Individuals who have financial literacy can utilize the various financial products offered in the market to make thorough investment or financial decisions. Thus, financial literacy must be a universal necessity and fundamental right, instead of privilege of limited people who have better financial knowledge access [17].

Students' fintech knowledge levels are measured to know the current state of knowledge between them and their willingness to address fintech practices and models to improve the university syllabus. Academics can start initiatives to include fintech knowledge and equip students with various considerations; students can currently acquire knowledge about fintech in the classroom. There is a short course so that students can understand the basics of fintech and restructure the current syllabus to fit the requirements of fintech [18].

The study results [19] showed individual factors could significantly predict students' financial literacy. Students from business majors have a higher understanding of financial concepts than other majors. Individuals who previously took finance courses had better performance. The study's findings highlight that class in financial literacy can provide a broad understanding for students. The study showed that higher confidence in their financial knowledge led to better scores. Adding courses related to financial yearning can be an option in improving students' financial literacy.

Study related fintech peer-to-peer lending [20] conclude that majority of respondents are aware by using application of peer-to-peer lending. The results of study show that peer-to-peer lending company have succeeded in generating a well user experience and interface. Generally, respondents were pleased with quality of service which is offered by the company. Furthermore, the company have provided suitable protection and security. The accuracy and comprehensive information provided are adequate to make good decisions about the loan application.

Research conducted [9] shows that user trust in fintech services significantly influences user adoption attitudes. Moreover, perceived risks and perceived ease of use not affected user attitudes towards fintech service adoption. The research provides a wide-ranging view of the user attitudes

determinants by combining of fintech service trust using the technology acceptance model (TAM). According to the study results [21], perceived value, assurance, speed, perceived firm innovativeness and customer support has positive effect to fintech customer experience. Fintech companies should take part customer experience into company business models. The research models and hypotheses proposed in this study are:



Fig. 1. Proposed research model

H1: There is an influence of financial literacy on fintech knowledge.

H2: There is an influence of financial literacy on fintech service usage through fintech knowledge.

III. RESEARCH METHODS

This research uses explanatory approach that intends to describe the position of variables and the relationships between variables used in the study. Data collection was conducted through questionnaires using a five-point Likert type scale in this study. On the Likert scale, respondents were asked to provide five answer options, namely strongly disagree (1), disagree (2), neutral (3), agree (4), strongly agree (5). The research subjects are students at Bina Nusantara University, BINUS Business School Undergraduate Program. The objects of the study are Financial Literacy, Fintech Knowledge and Fintech Services Usage.

Financial literacy is a series of knowledge, behavior and skills of consumers and communities to manage finances better. Indicators measure Financial Literacy (FL): Understanding interest rates (UIR), Understanding inflation (UIN), Financial skill (FSK), Financial attitude (FAT), Management of finances (MOF), Risk and return in investing (RAR) [13]; [14]; [16]; [22].

Fintech knowledge is a level of knowledge in general related to fintech and its practices. Indicators measure Fintech Knowledge (FK): Knowledge about fintech practices (KFP), Knowledge about fintech terms (KFT), Knowledge from college (KCO), Familiarity in fintech (FIF), Awareness of fintech (AOF) [5]; [18].

Fintech Services Usage is a user experience of the services provided by fintech company platforms. Indicators measure Fintech Services Usage (FS): Access features (AFE), Service reliability (SRE), User's needs (UNE), User-friendly Application (APP), Trust (TRU), Intention (INT) [9]; [20]; [23].

Questionnaires that the respondent has returned are then conducted descriptive statistical testing and Structural Equation Modelling (SEM) test using SmartPLS. Data collection using questionnaires was conducted from January to March 2022. The number of respondents who filled out questionnaires was as many as 210 students. Respondents' profiles were categorized by gender and student's year at university. The number of respondents answered questionnaire questions based on gender, Men respondents 130 (61.90%), and Women respondents 80 (38.10%). The

number of respondents who answered questionnaire questions based on Student classification for first-year respondents or Freshman 151 students (71.90%), second-year respondents or Sophomore 32 students (15.24%), third-year respondents or Junior 14 students (6.67%), fourth-year and more respondents or Senior 13 students (6.19%).

The analytical research methods are descriptive and multivariate analysis. Analytical research method conducted descriptive analysis to get perceptions related to the variables of Financial Literacy, Fintech Knowledge and Fintech Services Usage. Multivariate analysis tests the proposed hypothesis; this study uses Structural Equation Modelling based on variance or Partial Least Square.

Structural Equation Modelling consists of outer model or measurement model and inner model or structural model. The outer model shows measurement of manifest variables or observed variables can represent latent variables. The outer model or measurement model used the reliability and validity tests. The validity test uses value of factor loadings criterion is > 0.60 , and the AVE or Average Variance Extracted value is > 0.50 . The Reliability tests use Cronbach Alpha values on each variable with values > 0.7 , and Composite Reliability values > 0.70 [24].

Structural models (inner models) show that estimates between latent variables or constructed constructs are accurate and robust. Structural model evaluation (inner model) includes a discussion of coefficient of determination or R^2 , Goodness of Fit or GoF test, predictive relevance test or Q^2 , and hypothesis test.

Test coefficient of determination (R^2) is a measure of predictive power, and the higher the coefficient determination value, the better the endogenous constructs' prediction. The criterion of the value R^2 for the value 0.75 means strong, the value 0.50 means moderate and the value of 0.25 means weak. Predictive relevance (Q^2) tests measure predictive capabilities in a research model. The value of predictive relevance, if greater than 0, indicates the prediction accuracy of the path model used in the study is acceptable. The value of predictive relevance is acquired by using procedure blindfolding and cross validated redundancy. The predictive relevance test criteria show that exogenous constructs have small predictive relevance are 0.02, for mediums is 0.15, and large is 0.35. [24]; [25].

The Goodness of Fit test (GoF) measures how well the theoretical structure stated by the data can represent reality. GoF values are calculated using the formula:

$$GoF = \sqrt{AVE \times R^2}$$

The GoF value is $0 \leq GoF \leq 1$ with the criteria used for value of small GoF is 0.10, for value of medium GoF is 0.25, and value of large GoF is 0.36 [26]. The hypothesis test uses an alpha of 5% or p-value 0.05. The criterion of hypothesis zero (H_0) is rejected if the p-value < 0.05 .

IV. RESEARCH RESULTS

Data collected through questionnaires are processed using statistical software SmartPLS to perform structural equation modelling analysis. The outer model or measurement model conducted test of validity and reliability. The test of validity uses the factor loadings value criteria on each indicator greater than 0.60. The results of validity test

for factor loadings Financial Attitude (FAT) is 0.346, Management of finances (MOF) is 0.449 and Knowledge from college (KCO) is 0.438. Indicators FAT, MOF, and KCO have value of loading factors less than 0.60, so that indicators is not included in the following analysis. The research model was re-analysed after several indicators with factor loadings values of less than 0.60 were excluded. Factor loadings that meet the criteria in the validity test are presented in Table 1.

TABLE I. FACTOR LOADINGS

Indicator	Factor-Loadings	Indicator	Factor-Loadings	Indicator	Factor-Loadings
UIR	0.75	KFT	0.88	UNE	0.82
UIN	0.74	FIF	0.85	APP	0.86
FSK	0.73	AOF	0.80	TRU	0.86
RAR	0.68	AFE	0.82	INT	0.79
KFP	0.89	SRE	0.82		

The average variance extracted or AVE value for all variables is > 0.50 , then it can be concluded that variables used in the research are valid. AVE value data is presented in Table 2 as follows.

TABLE II. AVERAGE VARIANCE EXTRACTED

Variable	Average Variance Extracted
Financial Literacy	0.53
Fintech Knowledge	0.73
Fintech Services Usage	0.69

Reliability tests use Cronbach's Alpha measurements, with the value criteria being more than 0.70. Based on the data results in Table 3, all variables has Cronbach's Alpha value above 0.70 meets the assessment criteria for internal consistency.

TABLE III. RELIABILITY TEST

Variable	Cronbach's Alpha
Fintech Services Usage	0.91
Fintech Knowledge	0.88
Financial Literacy	0.70

In the discussion of inner model or structural model conducted Goodness of Fit (GoF) test, predictive relevance test (Q^2), coefficient of determination (R^2), and hypothesis test. Based on the data results presented in Fig. 2, the value of R^2 for the variable Fintech Knowledge (FK) is 0.345. The results interpreted that the Fintech Knowledge (FK) variable that can be explained by Financial Literacy (FL) variability of 34.5% for the rest is described by another variables that exist outside the research. The R^2 value for variable Fintech Services Usage (FS) is 0.454. The results are interpreted that the Fintech Services Usage (FS) variable that can be explained by the variability of Fintech Knowledge (FK) of 45.4% for the rest is described by another variables outside the research.

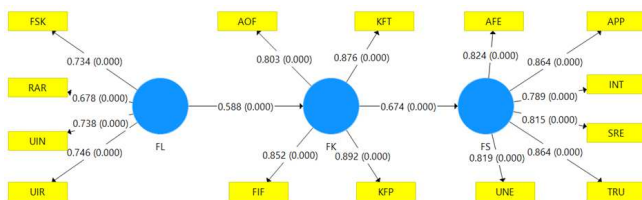


Fig. 2. Data processing results using SmartPLS

Using (1) obtained the GoF value is 0.51, according to the criteria used, then the GoF value > 0.36 therefore the evaluation for overall measurement model has a large result. The result of the data presented in Table 4 obtained the value of Q^2 is 0.29. The criteria used for Q^2 values show that exogenous constructs have medium predictive relevance ($0.15 < Q^2 < 0.35$).

TABLE IV. TEST OF PREDICTIVE RELEVANCE (Q^2)

Description	Fintech Services Usage
SSO	1,260.00
SSE	894.916
$Q^2 = 1 - SSE/SSO$	0.29

The hypothesis test uses significance level $\alpha = 5\%$ so that the proposed hypothesis is significant if the p-value < 0.05 . Hypothesis test 1 has a path coefficient of 0.59, and the p-value is $0.00 < 0.05$. So, the conclusion for hypothesis test 1 is a positive and significant influence of Financial Literacy on Fintech Knowledge. Hypothesis test 2 has path coefficient 0.40 and p-value of $0.00 < 0.05$. The conclusion obtained in the hypothesis 2 test, namely the influence of Financial Literacy on Fintech Services Usage through Fintech Knowledge, is positive and significant.

TABLE V. HYPOTHESIS TEST RESULTS

Description	Path-Analysis	Path-Coefficient	P-Value	Decision
Hypothesis 1	FL $\rightarrow$ FK	0.59	0.00	Positive and Significant
Hypothesis 2	FL $\rightarrow$ FK $\rightarrow$ FS	0.40	0.00	Positive and Significant

Based on the results of hypothesis test obtain in Table 5, the decision for the proposed hypothesis is to accept hypothesis one (H1) with the conclusion that there is an influence of financial literacy on fintech knowledge. Accept hypothesis two (H2) with the conclusion that financial literacy influences fintech service usage through fintech knowledge.

V. CONCLUSION

The services provided by the fintech industry in Indonesia have experienced rapid growth supported by affordable mobile phone and computer prices and also technology advances in internet services. In financial services, the assistance offered by fintech companies can represent services such as banks and other financial institutions. Increasingly advanced digital technology devices support the rapid development of fintech. Local knowledge and level of financial literacy become important factors for successful design and also implementation of fintech services in emerging countries [4].

Financial literacy has significant and positive impact on fintech knowledge. Fintech knowledge support is needed so that the effect of financial literacy on fintech services usage becomes positive and significant. The results of this research are consistent with previous study [23], which concluded the need for awareness improvement for an acceleration of fintech services usage, in addition to financial literacy, is needed to support consumers in using fintech applications [17]; [20]. Fintech knowledge among

students supports the improvement of Fintech Services Usage [5]; [18].

Financial literacy in college students still needs to be improved; one way is to provide courses on fintech or include fintech discussions in the related course of study. High levels of financial literacy and supported by good fintech knowledge can make fintech services usage among college students increase so that it can ultimately encourage the development of the fintech industry. Further research can expand the reach of respondents in high school students to find out the effect of financial literacy on fintech services usage among teenagers.

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